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Cybersecurity

Fundamentals of Cybersecurity

CIA Triad (Confidentiality, Integrity, Availability)

The CIA Triad forms the backbone of cybersecurity principles. Confidentiality ensures data privacy, integrity maintains accuracy, and availability guarantees access when needed. Together, they define the core goals of secure systems.

Authentication & Authorization

Authentication verifies user identity, while authorization decides what resources they can access. Strong mechanisms like multi-factor authentication enhance security. These processes prevent unauthorized access and misuse of data.

Encryption Basics

Encryption converts data into unreadable code to protect it from attackers. Symmetric and asymmetric methods are widely used in securing communication. It ensures sensitive information remains safe during transmission.

Network & Application Security

Firewalls & Intrusion Detection Systems (IDS)

Firewalls act as barriers between trusted and untrusted networks. IDS monitors traffic for suspicious activity and alerts administrators. Together, they form the first line of defense against cyber threats.

Web Application Vulnerabilities (SQL Injection, XSS)

Applications often face attacks like SQL injection and cross-site scripting (XSS). These exploit coding flaws to steal or manipulate data. Secure coding practices and regular testing help prevent them.

Patch Management

Software vulnerabilities are often exploited by attackers. Patch management ensures timely updates to fix these weaknesses. It is a critical step in maintaining system security and reliability.

Types of Cyber Threats

Malware (Viruses, Worms, Ransomware)

Malware is malicious software designed to disrupt, damage, or steal data. Viruses attach to files, worms spread independently, and ransomware locks data for payment. Defenses include antivirus tools and regular updates.

Phishing & Social Engineering

Phishing tricks users into revealing sensitive information through fake emails or websites. Social engineering manipulates human psychology to bypass security. Awareness and training are key to preventing these attacks.

Denial of Service (DoS/DDoS)

DoS attacks flood systems with traffic, making them unavailable to users. DDoS amplifies this by using multiple compromised devices. Mitigation involves firewalls, traffic filtering, and robust network infrastructure.

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